The Value of Metadata in Digital Forensics

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***Abstract*— Metadata is not visible when viewing data in a number of forms such as a word document or an image. It is, however, an important consideration in the discovery of information for use in digital forensic investigations. Different types of documents and files have a number of formats and types of metadata, which can be used to discover the properties of a file, document or network activity. Moreover, Metadata is useful in many circumstances, where it can provide collaboration evidence of between groups of people, because some of them are not aware of which type of information is stored within their document. Thus, the digital forensics investigator can access to this hidden document information. In legal cases, the identification of relevant digital evidence is crucial for supporting the case, verification and an examination existing legal argument forms. In this work, we show how to use the different formats and types of metadata in order to validate the legal argument for relevant evidence.**

Keywords- Metadata; Digital Evidence; Digital Forensics; legal practitioner; Investigator.

I. INTRODUCTION

Electronic information often contains metadata that cannot be seen when viewing the information using the applications and tools that are conventionally associated with the file type. Metadata can, however, provide an enormous body of information to a digital investigator, also, it can be utilised to incriminate (or exonerate) the author, reviewer, owner or publisher of the document of file.

II. METADATA

Metadata can provide the investigator with a wealth of information on the files that are being investigated. Furthermore, the forensic investigator can use metadata to obtain information, for instance, the file's author, the date and time of creation, the number of times the file has been modified, including when the modification took places. Common programmes such as MS Office and OpenOffice.org contain features such as track changes and comments that provide a means for users to view the various alterations that have been made to the document. File System Metadata included in the category of file system metadata is information about file size, specific data units allocated and access to date or time stamps within the file. Metadata is contained in media files such as graphic interchange format (GIF), JPEG, and in music: MP3 and AAC and tagged image file Format. Metadata is included in document files such as Microsoft office, office open XML format (MS Office 2007), Open Office and Portable document format. Thus, metadata can be useful in the resolution of legal disputes, because it can be used as evidence for the proving or disproving of other evidence put forward in a court case. Additionally, metadata can be kept in various sites inside files. Investigators can glean information from this metadata that correlates to ownership and potential owners [1][2].

III. ADMISSIBILITY OF METADATA

The Information revealed by metadata may be very important to an investigator in discovering any changes or manipulation, and it helps legal practitioners to reach to their inferences about the case. An investigation may also fail when evidence is not validated, or subject to adequate scrutiny during the investigation. The different factors that might affect the validity of evidence are use of unsuitable tools, systems or application errors during the process of evidence collection, failure to report exculpatory evidence, misrepresentation of evidence, failure to identify pertinent evidence, and falsification of evidence leading to misdirection. Thus, the legal practitioner must understand how digital evidence is collected; and the relationship between the collection process and the corroboration of potential evidence [3].

IV. CONCLUSION

It can be seen from the above discussion that metadata can produce evidence that is not always readily apparent and the investigator must ensure the collection of such data and the validation of it for presentation in court. It is important that legal practitioners dealing with cases that involve metadata are made aware of the nature and value of the metadata to assist in the prosecution process.

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